

Date	03 July 2026
Team ID	SWTID-2026-4799
Team Size	5
Team Leader	Hari Bellamkonda
Team Member 1	Challa Sukumar
Team Member 2	Umesh CHALAPADHI
Team Member 3	Girish Kumar
Team Member 4	Vignesh A
Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Sample Project Documentation

Introduction

"Ibe Sample Project Lhunnrtation provides a comprehensive overview Of the Machine Learning—Based development methodology, Credit Card Approval Prediction System. It surnnurizes the project objectives, software amhitecture, invlenrtation approach, technologies used, and exp:cted Outcon-es. •mis doc-ument as a reference for understanding the overall design and execution Of the proposed intelligent creditcard am-moval prediction solution.

Project Overview

Project Title	Machine Iwarning—Based Cledit Card Approval Prediction System
Ihnain	Artificial Intelligence and Machine Learning
Application	Web-based Intelligent Prediction System
I-hmvanuning Language	
Framework	
Database	MySQL
Machirw Iæarning Libraries	Scikit-learn, mndas, NurnPy
Development "InLs	Visual Studio Code, Jupyter Notebock
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Project Objectives

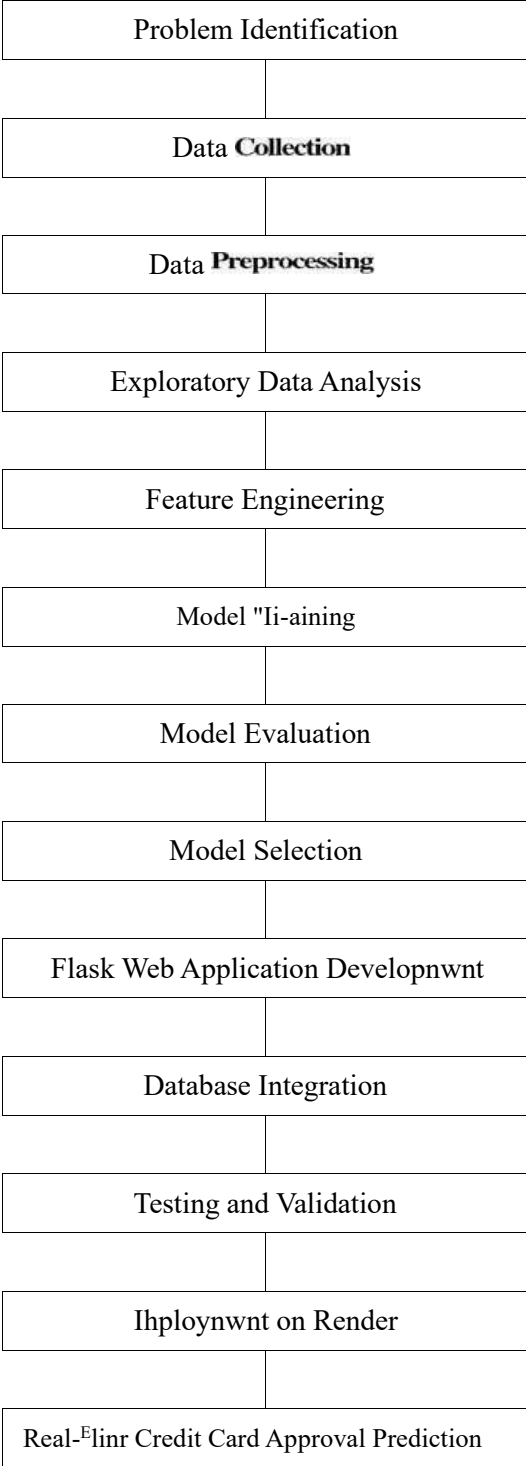
- Automate the credit card approval process.
- Reduce manual verification effort.
- Improve decision-making accuracy using Machine Learning.
- Provide real-time approval prediction.
- Develop an intuitive web application.
- Maintain secure applicant information.
- Support scalable deployment.
- Improve operational efficiency for financial institutions.

Major Modules

M(XIule	Description
t_Jsær Authentication	registration and login.
Applicant Managenent	Collects am.licant infornution.
Data Preprxwessing	data for prediction.
N'lzæhine l_ßarning mediction	medicts approval or rejection.
Database Management	Stoves applicant Rx-orris.
Result Dashboard	D)isplays prediction lesults.

Project Workflow

Project Features



Expected Outcomes

	Description
Accurate prediction	Improves approval decision accuracy.
Automation	Reduces manual processing effort.
Efficiency	Provides faster application processing.
Security	Protects applicant information.
Scalability	Supports future system expansion.
Maintainability	Simplifies future enhancements.

- Secure user authentication.
- Applicant profile management.
- Intelligent credit approval prediction.
- Multiple Machine Learning algorithms.
- Responsive web interface.
- Database integration.
- Prediction history management.
- Real-time decision support.

Technology Summary

Frontend	HTML-5, CSS3, JavaScript, Bootstrap 5
	Python,
Machine Learning	Scikit-learn, pandas,
Database	MySQL
Deployment	Render and Docker
Version Control	Git GitHub

Conclusion

The Machine Learning—Based Credit Card Approval Prediction System demonstrates how Artificial Intelligence can automate financial decision-making through accurate prediction, data management, and an intuitive web application. The project integrates modern Machine Learning techniques with technologies to deliver a scalable, reliable, and user-friendly solution for credit card approval prediction.